Calc I Lecture 5: Continuity and Vertical asymptotes

Let's revisit the definition of continuity:

Definition: A function f is continuous at a point a if $\lim_{x \to a} f(x) = f(a)$

intuition: a continuous function can be drawn without lifting the pen

Note: the above definition implicitly requires three things:

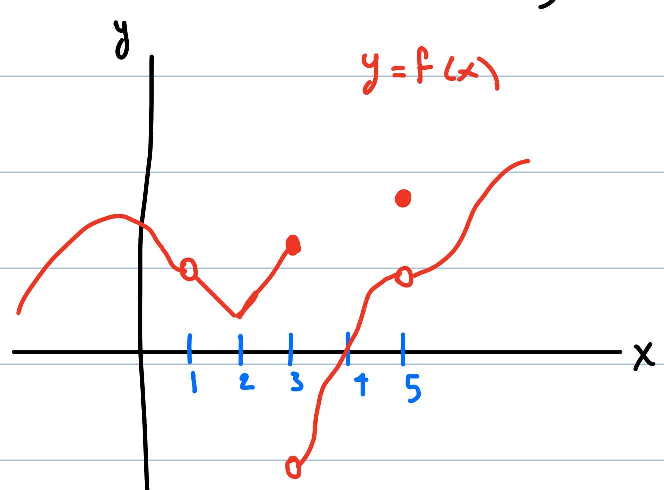
i) $f(a)$ is defined (a is in the domain of f)

ii) $\lim_{x \to a} f(x)$ exists

iii) $\lim_{x \to a} f(x) = f(a)$

In contrast, we say that f is discontinuous at a if it is defined near a but not continuous there (so one of the above conditions fails)

Example: Consider the graph of a function f shown below. Identify values at x where f is not continuous



• $x = 1$: $f(1)$ not defined $\Rightarrow$ condition i) fails.

• $x = 3$: i) $f(3)$ is defined

ii) $\lim_{x \to 3} f(x)$ DNE $\Rightarrow$ condition ii) fails

• $x = 5$: i) $f(5)$ is defined, ii) $\lim_{x \to 5} f(x)$ exists

iii) $\lim_{x \to 5} f(x) \neq f(5)$ $\Rightarrow$ condition iii) fails.

<u>Example</u> at what points are each of the following functions discontinuous?

a) $f(x) = \dfrac{x^2 - x - 2}{x - 2}$

b) $f(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$

c) $f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 2 & \text{if } x = 2 \end{cases}$

d) $f(x) = \lfloor x \rfloor$ (floor function)

Solutions:

a) $f(x) = \dfrac{x^2 - x - 2}{x - 2}$

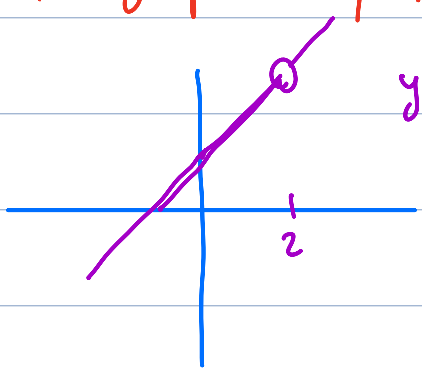
Remember: functions in our "library of functions" are continuous whenever they are defined

$f(x)$ is a rational function, which is in our "library"

$\Rightarrow$ continuous wherever it is defined

only discontinuous when not defined: $\boxed{x=2}$ (dividing by 0)

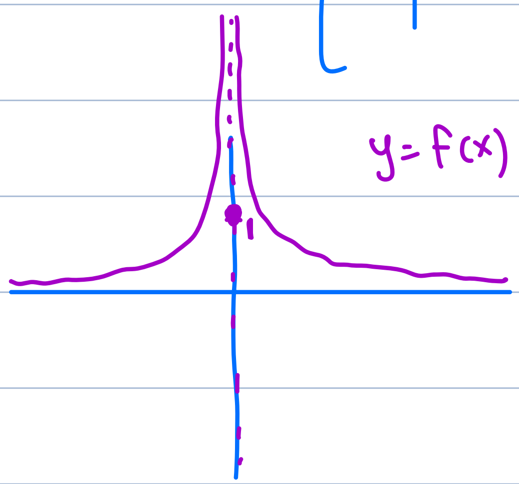
To graph $f(x)$, note $\dfrac{x^2-x-2}{x-2} = \dfrac{(x-\cancel{2})(x+1)}{x-\cancel{2}} = x+1$ when $x \neq 2$



b) $f(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$

this is only discontinuity because $\frac{1}{x^2}$ is continuous everywhere except 0

$\downarrow$



discontinuous at $x=0$

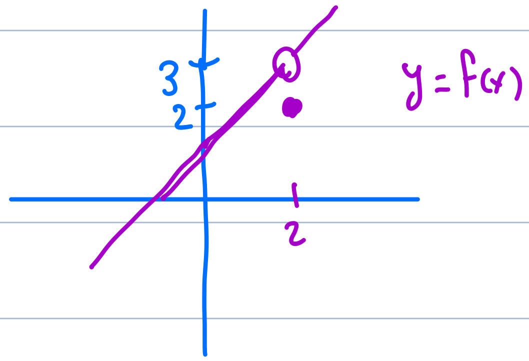
i) $f(0)$ is defined, $f(0) = 1$

ii) $\lim\limits_{x \to 0} \dfrac{1}{x^2} = \infty$ DNE

condition ii) fails

c) $f(x) = \begin{cases} \dfrac{x^2-x-2}{x-2} & \text{if } x \neq 2 \\ 2 & \text{if } x = 2 \end{cases}$

$$\frac{x^2-x-2}{x-2} = \frac{(x-\cancel{2})(x+1)}{x\cancel{-2}} = x+1 \quad \text{when } x \neq 2$$



i) $f(2)$ exists, $f(2)=2$

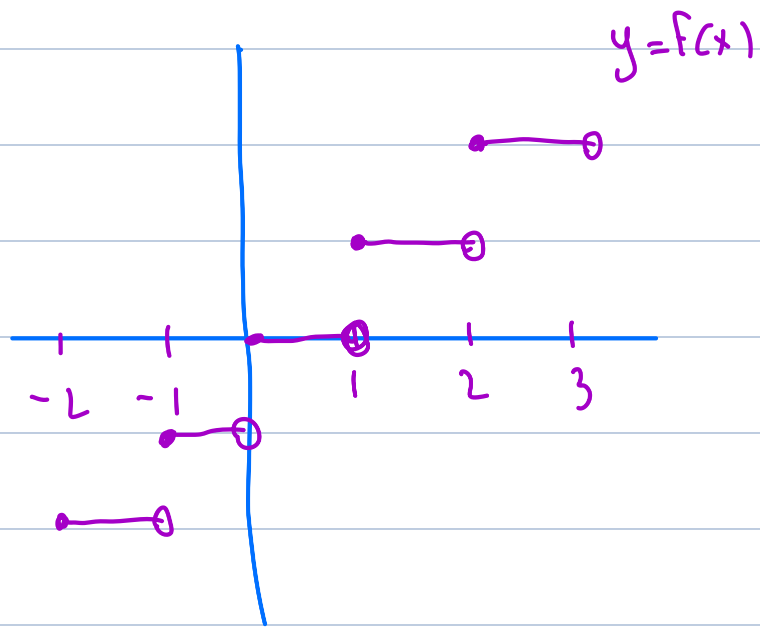
ii) $\lim_{x \to 2} f(x) = 3$

iii) $\lim_{x \to 2} f(x) \neq f(2) \Rightarrow$ condition iii fails

discontinuous at 2

continuous everywhere else.

d) $f(x) = \lfloor x \rfloor$



we've seen before:

$\lim_{x \to n} f(x)$ DNE

for any whole number n

condition ii) fails.

- discontinuous at every whole number
- continuous everywhere else

These functions exhibit different types of discontinuity:

a) & c) "removable discontinuity"

technical definition: $f(x)$ is discontinuous at a but $\lim_{x \to a} f(x)$ exists

b) "infinite discontinuity"

$f(x)$ has a vertical asymptote as $x \to a$

d) "jump discontinuity":

$$\left.\begin{array}{l} \lim_{x \to a^-} f(x) \text{ exist} \\ \lim_{x \to a^+} f(x) \text{ exist} \end{array}\right\} \text{but not equal}$$

* technically we can have other discontinuities too, e.g. $\sin\left(\frac{1}{x}\right)$

One-sided continuity

Just like limits, we can consider continuity from the left or from the right:

- A function $f(x)$ is continuous from the right at a if $\lim_{x \to a^+} f(x) = f(a)$
- A function $f(x)$ is continuous from the left at a if $\lim_{x \to a^-} f(x) = f(a)$

Example For each of the discontinuities in the previous example, is the function continuous from the right, the left, or neither?

a) neither ($f(a)$ is not defined)

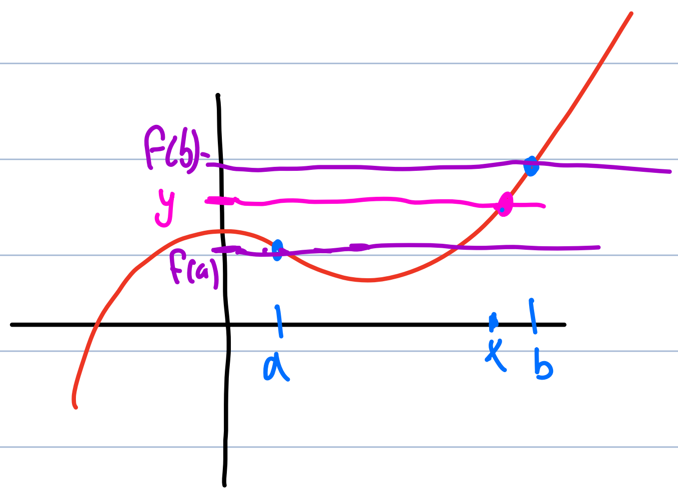
b) neither (both one-sided limits DNE)

c) neither (both one-sided limits do not equal $f(a)$)

d) continuous from the right at every whole number but not from the left

The Intermediate Value Theorem

Intuition: if $f(x)$ is continuous over $[a,b]$
it must traverse all values between $f(a)$ and $f(b)$
as x goes from a to b



Statement: if $f(x)$ is continuous over $[a,b]$
for every $f(a) \leq y \leq f(b)$
there exists some $a \leq x \leq b$ such that $f(x) = y$

Caution: this fails if f is not continuous.

Example: Use IVT to show that

$f(x) = x^3 - 2x^2 - x - 5$ must have a root.

Idea: want $x^3 - 2x^2 - x - 5 = 0$

find b where $f(b) > 0$ (e.g. $x = 10 \Rightarrow f(x) = 1000 - 200 - 10 - 5 > 0$)

find a where $f(a) < 0$ (e.g. $x = -10$ $f(x) < 0$)

$f(a) < 0 < f(b) \Rightarrow$ by IVT there is some x

where $a \le x \le b$

and $f(x) = 0$

(note $f(x)$ is continuous because it's a polynomial so IVT applies)

Question: does the same idea work for all cubic polynomials?

Example Prove that

$$\overbrace{3^x - x^5}^{F(x)} = \sqrt{2}$$

must have a solution for some x in $[\underset{a}{0}, \underset{b}{1}]$

Use IVT

$f(x) = 3^x - x^5$

$f(0) = 3^0 - 0^5 = 1 - 0 = 1$

$f(1) = 3^1 - 1^5 = 3 - 1 = 2$

$F(0) = 1 < \sqrt{2} < 2 = f(1)$

$\Downarrow$ IVT } IVT applies because $f(x)$ is continuous everywhere

There exists $0 < x < 1$

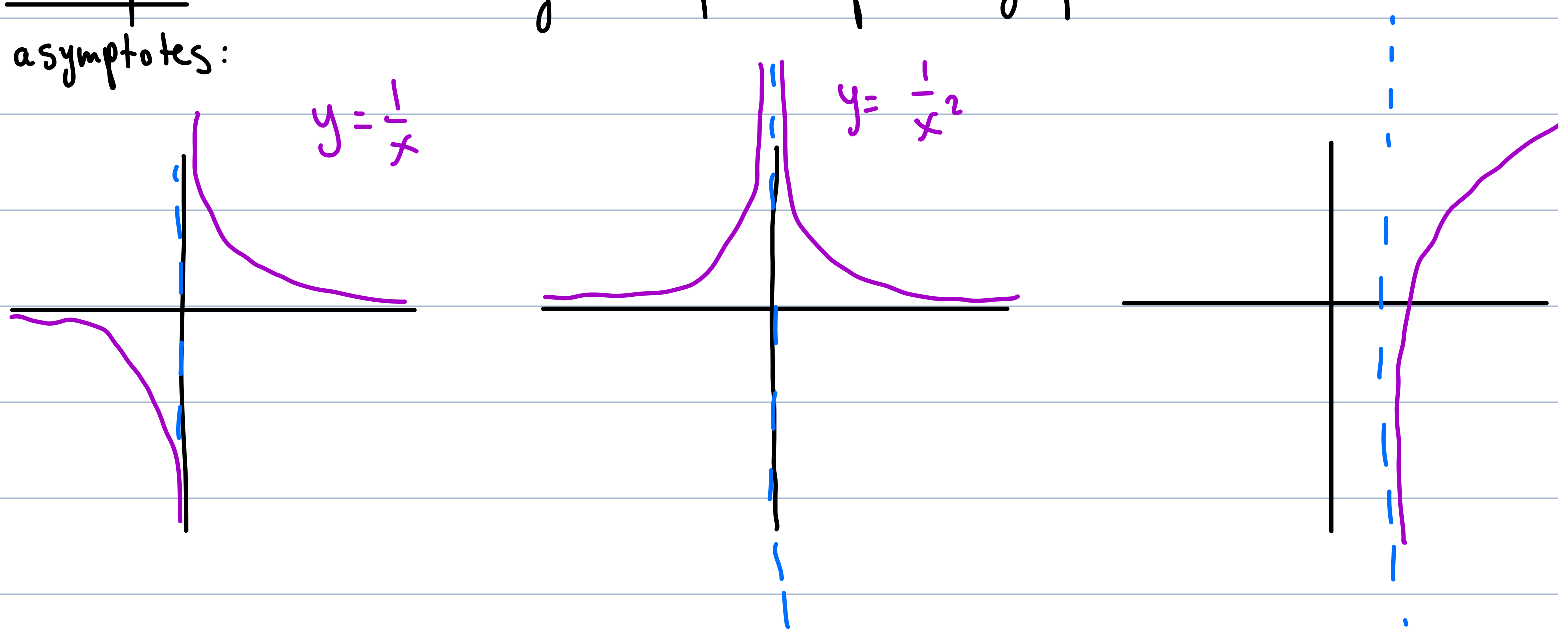
such that $f(x) = \sqrt{2}$

Asymptotes

We have two types of asymptotes:
vertical asymptotes and horizontal asymptotes

Definition: Given a function $f(x)$, we say that $x=a$ is a vertical asymptote of f if at least one of the one-sided limits is ∞ or $-\infty$. $\left(\lim_{x\to a^+} f(x) = \pm\infty \text{ OR } \lim_{x\to a^-} f(x) = \pm\infty\right)$

Example: the following examples depict graphs with vertical asymptotes:



Note: vertical asymptotes generally arise at places where we divide by 0. (or for $\ln(x)$)

More specifically:

if $\dfrac{f(x)}{g(x)}$ (suppose f, g continuous)

$g(a)=0$ and $f(a)\neq 0 \Rightarrow \left|\dfrac{f(x)}{g(x)}\right| \to \infty$ as $x\to a$

Example: Identify all the vertical asymptotes of $f(x)=\tan(x)$

$$f(x)=\tan(x)=\frac{\sin(x)}{\cos(x)}$$

vertical asymptotes occur when $\cos(x)=0$

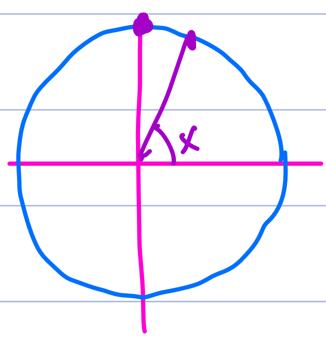
$$x=\left\{\frac{\pi}{2}+n\pi \text{ where } n \text{ is a whole number}\right\}$$

$$=\left\{\frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots, -\frac{\pi}{2}, -\frac{3\pi}{2}, -\frac{5\pi}{2}, \dots\right\}$$

Followup: $\displaystyle\lim_{x\to\frac{\pi}{2}^-} \tan(x) = \boxed{\infty}$

assume $x<\frac{\pi}{2}$

$\pi/2$



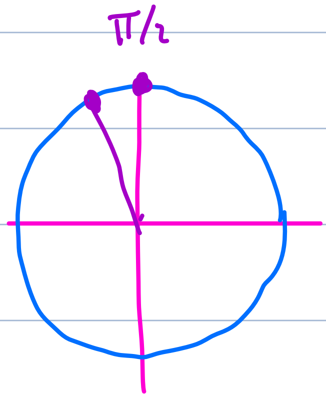
$$\tan(x) = \frac{\sin(x)\}+}{\cos(x)\}+} = +$$

as $x\to\frac{\pi}{2}^-$, $\sin(x)\to\sin\left(\frac{\pi}{2}\right)=1>0$

$\cos(x)>0$

$$\lim_{x \to \frac{\pi}{2}^+} \tan(x) = \boxed{-\infty}$$

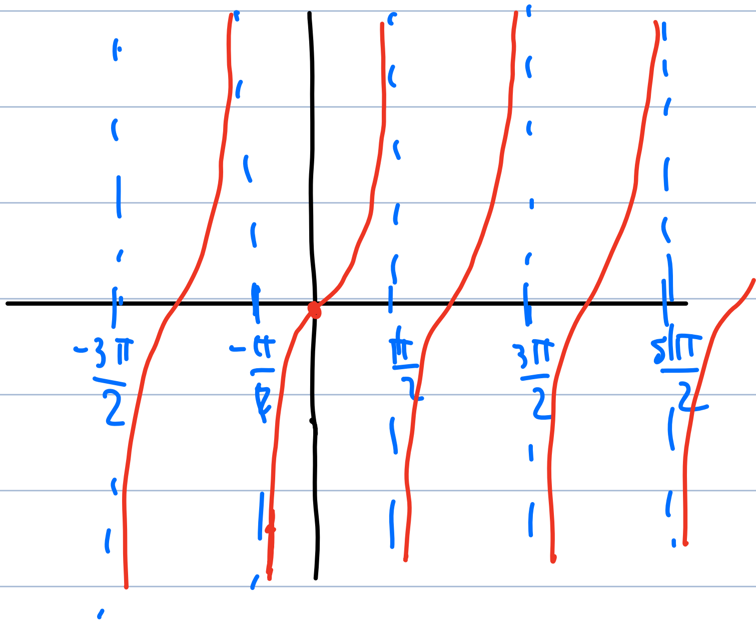
assume $x > \frac{\pi}{2}$



$$\tan(x) = \frac{\sin(x)}{\cos(x)} \begin{matrix} + \\ - \end{matrix} = -$$

as $x \to \frac{\pi}{2}^+$, $\sin(x) \to \sin\left(\frac{\pi}{2}\right) = 1 > 0$

$\cos(x) < 0$



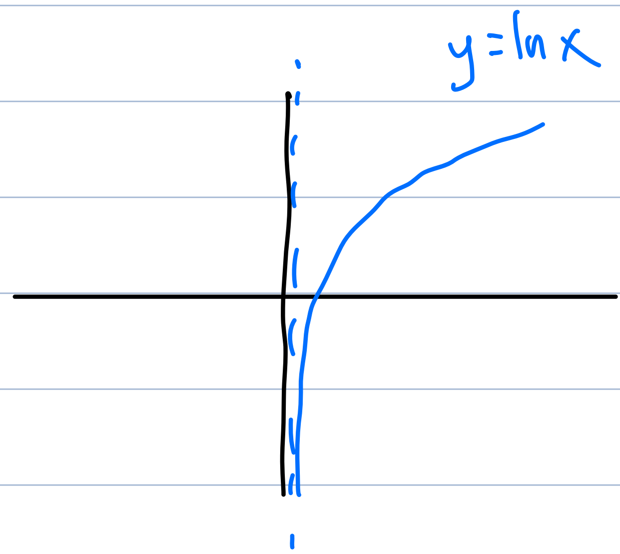
<u>Example</u>: a) identify all vertical asymptotes of the function $f(x) = \dfrac{\ln(x)}{x-2}$

b) Find the one-sided infinite limits at each asymptote

a)

• $x=2$ (division by 0) note: $\ln(2) \neq 0$

• $x=0$ (asymptote of $\ln x$)



b) • $\lim\limits_{x \to 2^-} \dfrac{\ln(x)}{x-2} =$

as $x \to 2^-$, $x < 2 \Rightarrow x-2 < 0$

$\ln(x) \to \ln(2) > 0$

$\dfrac{\ln(x) \}+}{x-2 \}-} = -$

$\lim\limits_{x \to 2^-} \dfrac{\ln(x)}{x-2} = \boxed{-\infty}$

inferior approach: plugging on some $x<2$ to find sign. e.g. plug in $x = 1.5$
(not full credit)

$$\lim_{x \to 2^+} \frac{\ln(x)}{x-2} =$$

as $x \to 2^+$, $x > 2 \Rightarrow x - 2 > 0$

$\ln(x) \to \ln(2) > 0$

$$\left.\begin{array}{l}\ln(x)\\ x-2\end{array}\right\} \frac{+}{+} = + \Rightarrow \boxed{\lim_{x \to 2^+} \frac{\ln(x)}{x-2} = \infty}$$

- $\lim_{x \to 0^-} \frac{\ln(x)}{x-2}$ = undefined (ln undefined for negative inputs)

$$\lim_{x \to 0^+} \frac{\ln(x)}{x-2} =$$

as $x \to 0^+$ $\ln(x) \to -\infty$

$x - 2 \to 0 - 2 = -2$

$$\left.\begin{array}{l}\ln(x)\\ x-2\end{array}\right\} \frac{-}{-} = + \Rightarrow \boxed{\lim_{x \to 0^+} \frac{\ln(x)}{x-2} = \infty}$$

$y = \frac{\ln(x)}{x-2}$

How to evaluate infinite limits? sign analysis!

Key idea: if we want to evaluate infinite limit

$$\lim_{x \to a} \frac{f(x)}{g(x)}$$

(suppose f and g continuous)

i. show $g(a) = 0$ but $f(x) \neq 0$.

ii. perform a sign analysis to see:

- if as $x \to a^+$, $\frac{f(x)}{g(x)}$ is positive when $x > a$ (and near a)

then $\lim_{x \to a^+} \frac{f(x)}{g(x)} = \infty$

- if as $x \to a^+$, $\frac{f(x)}{g(x)}$ is negative when $x > a$ (and near a)

then $\lim_{x \to a^+} \frac{f(x)}{g(x)} = -\infty$

Example: Suppose $f(x) = \dfrac{e^x}{(x-2)^3}$.

a) Find the infinite limit $\lim_{x \to 2^+} f(x)$

b) Find the infinite limit $\lim_{x \to 2^-} f(x)$

c) What can you say about $\lim_{x \to 2} f(x)$?

a) note numerator e^x at $x=2$: $e^2 \neq 0$
denominator $(x-2)^3$ at $x=2$: $0^3 = 0$ } goes to infinity

use sign analysis to find $+$ or $-$

$$\lim_{x \to 2^+} \frac{e^x \;\}\; e^2 \quad +}{(x-2)^3 \;\}\; +}$$

considering $x > 2 \Rightarrow x-2 > 0 \Rightarrow (x-2)^3 > 0$

$$\frac{+}{+} = +$$

$$\Rightarrow \lim_{x \to 2^+} \frac{e^x}{(x-2)^3} = \infty$$

b) if $x < 2 \Rightarrow x-2 < 0 \Rightarrow (x-2)^3 < 0$

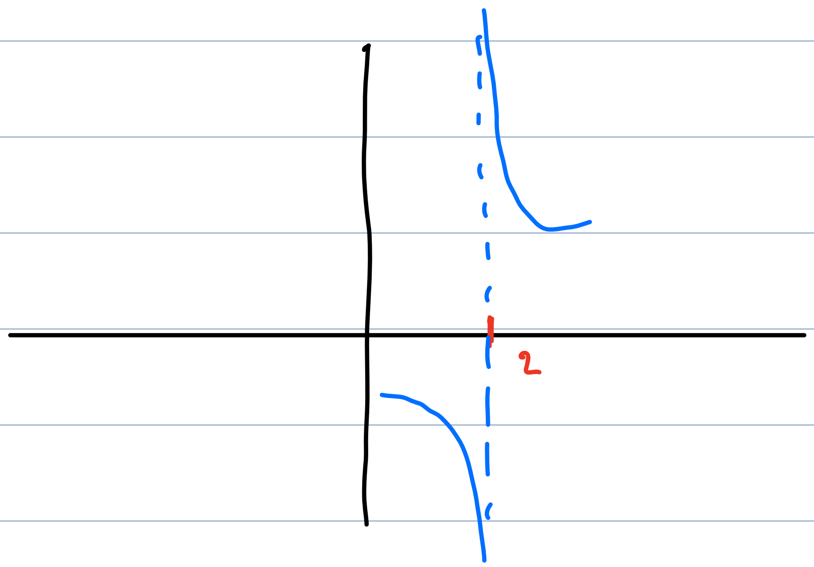
$$e^2 > 0$$

$$\frac{e^x}{(x-2)^3} \quad \frac{+}{-} = - \quad \Rightarrow \lim_{x \to 2^-} \frac{e^x}{(x-2)^3} = -\infty$$

c) $\lim_{x \to 2} f(x)$? neither ∞ nor $-\infty$, DNE

sketch behavior of function near $x=2$:

$$f(x) = \frac{e^x}{(x-2)^3}$$



Infinite limit laws table

informal	formal (let c be a positive number)
addition	
$\infty + \infty = \infty$	if $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = \infty$, then $\lim_{x\to a} (f(x)+g(x)) = \infty$
$-\infty + (-\infty) = -\infty$	⋮
$\infty + (-\infty) =$ indeterminate (it depends)	if $\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = -\infty$, then $\lim_{x\to a} (f(x)+g(x)) =$ indeterminate
$c + \infty = \infty$ $-c + \infty = \infty$	if $\lim_{x\to a} f(x)$ is finite and $\lim_{x\to a} g(x) = \infty$, then $\lim_{x\to a} (f(x)+g(x)) = \infty$
$c + (-\infty) = -\infty$ $-c + (-\infty) = -\infty$	

multiplication

$\infty \cdot \infty = \infty$

$\infty \cdot (-\infty)$

$= -\infty$

$(-\infty) \cdot (-\infty)$

$= \infty$

$(c \neq 0)$

$c \cdot \infty = \infty$

$c \cdot (-\infty) = -\infty$

$-c \cdot \infty = -\infty$

$-c \cdot (-\infty) = \infty$

$0 \cdot \infty =$ indeterminate

if $\lim_{x \to a} f(x) = 0$ and $\lim_{x \to a} g(x) = \infty$, then

$\lim_{x \to a} \left(f(x) \cdot g(x) \right) =$ indeterminate division

$\frac{c}{0^+} = \infty$	if $\lim_{x \to a} f(x) = c > 0$ and $\lim_{x \to a} g(x) = 0$ and g is positive near a then $\lim_{x \to a} \left(f(x)/g(x) \right) = \infty$
$\frac{c}{0^-} = -\infty$ $\frac{-c}{0^+} = -\infty$ $\frac{-c}{0^-} = \infty$	if $\lim_{x \to a} f(x) = c > 0$ and $\lim_{x \to a} g(x) = 0$ and g is negative near a then $\lim_{x \to a} \left(f(x)/g(x) \right) = -\infty$
$\frac{0}{0} =$ indeterminate	
$\frac{\infty}{c} = \infty$ $-\frac{\infty}{c} = -\infty$ $\frac{\infty}{-c} = -\infty$ $\frac{-\infty}{-c} = \infty$	

$$\frac{\infty}{0^+} = \infty$$

$$\frac{\infty}{0^-} = -\infty$$

$$\frac{-\infty}{0^+} = -\infty$$

$$\frac{-\infty}{0^-} = \infty$$

$$\frac{\infty}{\infty} =$$

indeterminate

$$\frac{\pm c}{\pm\infty} = 0$$

$$\frac{0}{\pm\infty} = 0$$